

AI/ML Projects & Interview Preparation Handbook

Knowledge Base for AI Resume Analyzer

1. Introduction to Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence, while Machine Learning (ML) allows systems to learn patterns from data. AI powers applications such as recommendation systems, autonomous vehicles, virtual assistants, fraud detection and healthcare analytics.

2. Deep Learning and Large Language Models

Deep Learning is a subset of ML that uses neural networks with multiple layers to solve complex problems such as image recognition and natural language processing. Large Language Models (LLMs) like GPT, Llama and Mistral generate human-like text and are used in chatbots, document analysis and coding assistants.

3. Resume-Worthy AI & ML Projects

Build projects such as Resume Analyzer, Image Classifier, Sentiment Analysis, Face Recognition Attendance System, Traffic Prediction, Chatbot using RAG, OCR Document Reader, Recommendation System and Plant Disease Detection. Clearly explain datasets, preprocessing, model architecture, evaluation metrics and deployment.

4. Interview Preparation

Prepare for technical interviews by reviewing Python, SQL, Data Structures, Machine Learning concepts, CNNs, APIs, Git and system design basics. Practice coding problems, explain previous projects confidently and revise fundamental concepts before interviews.

5. Behavioral and HR Interviews

Behavioral interviews evaluate teamwork, leadership, communication and problem-solving skills. Use the STAR (Situation, Task, Action, Result) framework to answer questions. Demonstrate curiosity, willingness to learn and examples of overcoming challenges.

6. Career Roadmap and Best Practices

Create a GitHub portfolio, publish projects, participate in hackathons, contribute to open-source software and continuously improve technical skills. Learn cloud platforms, Docker, databases and deployment techniques. Build a habit of documenting projects professionally and tailoring resumes for each opportunity.